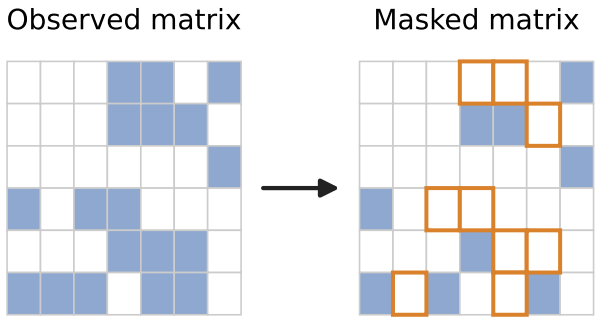
Controlled masking

PBMC 3K (human) + Paul15 (mouse)



genes × cells

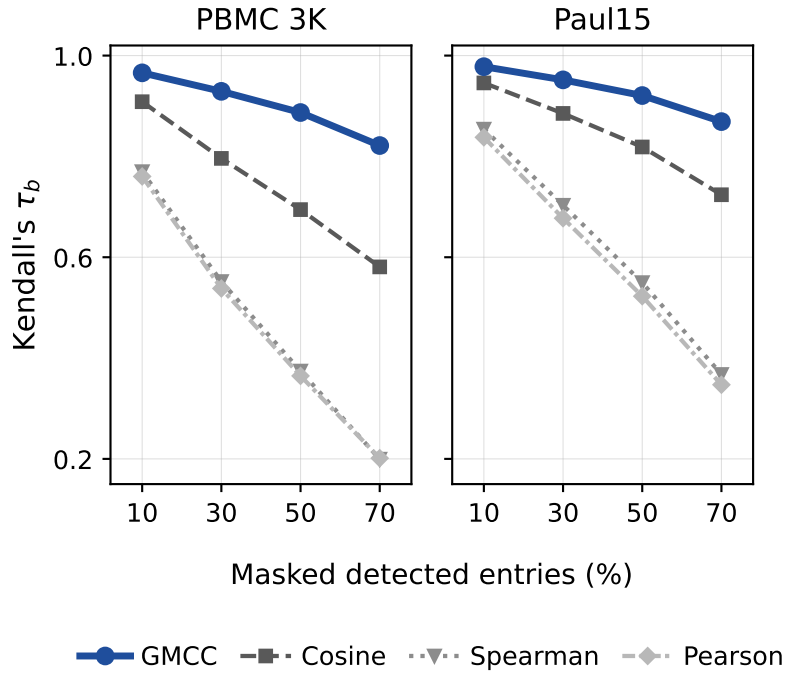
Detected entries randomly masked to zero (10–70%)

20 independent realisations per condition

GMCC | Pearson | Cosine | Spearman

Gene-pair rank robustness

Higher is more robust



GMCC also showed low magnitude change at moderate–severe masking

Main findings

Highest rank robustness
GMCC in both datasets

Recompute the global mean
new μ from each masked matrix

Different hub detection profiles
GMCC → lower-detection genes
Cosine → broadly detected genes

Enrichment varies by estimator and dataset
PBMC 3K: GMCC → immune processes
Paul15: Cosine → NK-cell processes

Robustness ≠ biological accuracy